



# Rajeev Mutyalu

Lead Software Architect • Production Technology Architect • Engineering & R&D Leader

- ★ Selected Production Technology Credits: 1917 • RRR • Mufasa: The Lion King
- Former Technicolor R&D Supervisor • Technicolor Fellowship Network Associate Member
- 🇬🇧 UK Skilled Worker Visa • Existing Sponsorship in Place • Available Immediately

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## Executive Summary

**Lead Software Architect and former Technicolor R&D Supervisor** with 20+ years of experience across VFX production technology, software architecture, product engineering and global engineering leadership. Proven track record designing and evolving production-critical platforms, proprietary applications and DCC-integrated systems across Astra Studios, Technicolor Group and MPC Film.

Experience spans production-grade OpenUSD architecture, global review platforms, modular Nuke ecosystems, Editorial Turnover / Conform / I/O systems, ShotGrid integrations, render and media infrastructure, studio automation and emerging AI production platforms. Led distributed engineering teams across London, Montreal and Bengaluru, establishing engineering standards and translating production requirements into scalable, maintainable technical systems. Combines long-term architectural thinking with hands-on engineering and deep understanding of real production workflows across major productions including [1917 \(Academy Award for Best Visual Effects\)](#), [RRR](#), [Mufasa: The Lion King](#), [Back in Action](#), [Spaceman](#), and [Prehistoric Planet](#). Associate Member of the Technicolor Fellowship Network Committee and recipient of 9+ studio leadership & excellence awards.

## Technical Arsenal & Core Competencies

Platforms

Product & Platform Architecture

- Enterprise VFX Platforms:** ReviewTool (Real-Time 4K+ Media Caching & Multi-Facility Sync), ProNuke Modular Shot Assembly, Meshroom Photogrammetry Integration.
- Scalable Media Systems:** High-throughput media playback, distributed cache synchronisation, RV integration, version-controlled pipeline hubs.

DCC & APIs

VFX Production Systems & DCC APIs

- DCC APIs:** Nuke C++/Python API, Maya Python API, Hiero / Nuke Studio Python API, 3D Equalizer, Silhouette.
- Production Tracking:** ShotGrid (Shotgun) Python API & Webhooks, schema customizations, automated task progression.

Leadership

Engineering Leadership & Delivery

- Distributed Teams:** Leading R&D Software Engineering, Development, and TD teams across London, Montreal, and Bengaluru.
- Engineering Standards:** Code review practices, architectural stewardship, Agile/Scrum delivery, bridging creative supervisors and technical developers.

OpenUSD

OpenUSD Modular Pipelines

- USD Architecture:** OpenUSD (USD Python API, Composition Arcs, Layer Stacking, Edit Targets, Sublayering), asset & shot schemas for full-CG and live-action VFX.
- Colour & Timeline Standards:** OpenColorIO (OCIO), ACES Colour Pipeline, OpenTimelineIO (OTIO) sequence interchange.

Core Systems

Core Software & Systems Architecture

- Languages & UI:** Python 3.x (Expert), PyQt / PySide (UI/UX Engineering), C++, Java, Objective-C, Cocoa, Shell Scripting.
- Infrastructure:** Docker, Linux/Unix environments, REST APIs, FastAPI, Git, Tractor Render Farm management & dispatching.

Conform & I/O

Editorial Turnover, Conform & I/O

- Turnover Ingest:** Package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero / Nuke Studio conform.
- Colour & Plates:** CDL grade extraction, shot LUT application, frame range validation, automated USD plate sublayering.

Standardisation

Cross-Studio Platform Standardisation

- Multi-Facility Toolsets:** RPM Unification across MPC, MR.X, and The Mill, standardising compositing and tracking toolsets.
- Shared UI/UX:** Reusable PyQt components, cross-studio software distribution, unified operational workflows.

GenAI & Automation

AI Platforms & Automation

- Generative Video Integration:** Studio.AI platform, ComfyUI node workflows, character identity consistency, ACES colour conforming.
- Event Automation:** n8n workflow automation hubs, inbound/reverse webhooks, automated show onboarding, smart plate validation.

Agentic & MCP

Agentic AI & MCP Systems

- Model Context Protocol (MCP):** Custom MCP server engineering, JSON-RPC 2.0 protocol, tool & resource schemas, pipeline API binding.
- Agentic Tooling:** Claude Code (CLI), OpenAI Codex, Google Antigravity, subagent orchestration, task decomposition.

Private AI

On-Premise Private AI & Security

- Local Inference:** Ollama, air-gapped studio deployments, local GGUF / AWQ model quantization, context window optimisation.
- IP Protection:** Secure internal AI endpoints preventing studio intellectual property leakage to public APIs.

## Professional Experience

CURRENT LEADERSHIP ROLE

Sep 2025 – Present

### Lead Software Architect

Astra Studios • London, UK

Focus: Production Technology, OpenUSD, VFX Platforms & Engineering Leadership

- **Engineering & Technical Leadership:** Lead Development and TD teams, defining production technology architecture and engineering standards while working directly with artists, production, supervisors and leadership to translate creative requirements into scalable technical systems.
- **Production Platform & OpenUSD Architecture:** Architected an end-to-end production-grade OpenUSD architecture connecting asset and shot workflows across DCCs (Maya, Nuke) and departments for full-CG and live-action VFX.
- **Editorial Turnover, Conform & I/O:** Engineered the studio's Editorial Turnover, Conform & I/O pipeline from scratch, integrating turnover package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero conform, CDL grades, shot LUTs, and USD plate sublayering.
- **Studio AI GenAI Platform Workflows:** Integrated generative AI workflows into studio infrastructure, including Studio AI prompt-to-pipeline automation, ComfyUI node workflows, and artist-facing controls with ACES colour conforming.
- **Zero-Touch Studio Automation:** Orchestrated zero-touch n8n studio automation hubs using inbound and reverse webhooks for automated show onboarding, smart plate validation, and ShotGrid metadata injection.

### R&D Supervisor

Mar 2023 – Feb 2025

Technicolor Group • London, UK

- **Global Review Platform Architecture:** Spearheaded the architecture and productionisation of ReviewTool, a studio-wide VFX review platform with real-time 4K+ media caching, multi-facility synchronisation and RV integration supporting distributed production workflows across global studios.
- **ProNuke Application Ecosystem:** Architected and engineered a modular Nuke production ecosystem spanning automated shot assembly, sequence-level publishing and batch render orchestration across global artist teams.
- **Meshroom Photogrammetry in Production:** Productionised Meshroom into matchmove and tracking workflows, cutting camera extraction turnaround from days to hours with fast switching between Meshroom, Maya, and 3D Equalizer.
- **Global R&D Leadership:** Led globally distributed R&D software engineering teams across London, Montreal, and Bengaluru supporting Compositing, Roto, Paint Prep, Camera Tracking, and Rotoanim departments.
- **Conform Ingest Automation:** Supervised Conform Ingest workflows using Hiero and Nuke Studio, automated CDL and LUT grade processing, and ShotGrid frame range validation.

### R&D Supervisor

Jan 2022 – Feb 2023

Technicolor Group • Bengaluru, India

- **2D Software Engineering:** Headed 2D software engineering and pipeline development across Bengaluru and Montreal hubs.
- **Review Platform Prototyping:** Architected core compositing automation tools and early prototypes of the studio VFX review platform (ReviewTool).
- **Agile Execution:** Mentored engineers, established Agile and Scrum sprint frameworks, and aligned dev priorities with production executives.

### Lead Pipeline Software Developer

Jul 2017 – Dec 2021

MPC Film • Bengaluru, India

- **Landmark Production Pipelines:** Led core 2D pipeline solutions across Academy Award-winning films and global tentpole productions (1917, RRR, The Lion King).
- **Cross-Studio Platform Standardisation:** Spearheaded RPM Unification across MPC, MR.X and The Mill, consolidating and standardising software toolsets and UI/UX across compositing and tracking workflows to improve technical consistency across multiple studio environments.
- **Nuke Bridge Hub:** Developed Nuke Bridge integrating Tessa (AMS), ShotGrid (PMS), and Tractor (Render Farm), establishing a version-controlled, high-throughput pipeline.
- **Render Dispatcher:** Engineered the Nuke Render Dispatcher with automated render order traversal and Reserve & Commit dispatch mechanics.

### Team Lead – Technology & R&D Production

Jul 2012 – Jun 2017

Technicolor • Bengaluru, India

- **Media Engines:** Architected macOS AVFoundation and QuickTime media processing engines (Channel Mapper, Frame Rate Converter, Compliance Edit Lists).
- **Team Management:** Managed an engineering team of 7 developers across C++, Objective-C, Cocoa, and Java, optimizing production compression and authoring pipelines.
- **Hardware Compliance:** Served as primary technical contact for player manufacturers, resolving Blu-ray authoring framework and playback compatibility issues.

### Senior Software Engineer – Technology

May 2008 – Jun 2012

Technicolor • Bengaluru, India • Los Angeles, USA • Hannover, Germany

- **Disney Collaboration (Los Angeles):** Collaborated on-site with Disney engineering in Los Angeles to architect the Fast Menu Load (FML) template and integrate Generation 2 Bootloaders (G2B).
- **World First 3D Blu-ray (Hannover):** Engineered core 3D capabilities into the BD-J Authoring framework at Technicolor R&I in Hannover, Germany, as a key contributor to the world's first released 3D Blu-ray disc.
- **Founding Engineer:** Founding software engineer at Technicolor India, building runtime alignment engines ('Graphics Adjuster') and studio authoring templates.

### Application Engineer & Game Programmer

Sep 2006 – Apr 2008

e4e & Dhruva Interactive • Bengaluru, India

- Ported mobile game titles to Nokia, Motorola, Samsung, and Sony Ericsson devices for T-Mobile and Verizon; engineered touchscreen interfaces for early mobile OS platforms.

## Honours, Awards & Fellowship



### Technicolor Fellowship Network

Associate Member • Technicolor Group

2015 - 2025

"Awarded for production environment expertise, strong customer relationships, and deep collaboration with Research & Innovation—delivering a rare mix of creativity, pragmatism, and architectural stewardship."

— Fellowship Nomination Committee • Mar 2015

Fellowship Committee Election   R&I Collaboration  
Architectural Leadership



### Studio Leadership & Excellence

2008 - 2024

9+ Honours • Technicolor, MPC Film & e4e



Value Champions & Best Employee of the Month (Aug 2019) —  
Technicolor FEV



Performer of the Quarter • Team Lead (Oct 2016) — Technology  
R&D Leadership



Performer of the Quarter (2011, 2012, 2014) — Disney  
collaboration & pipeline innovation



Extra Miler & Executive Team Awards (2011, 2013) — DVD/Blu-ray software



## Recent AI Professional Development

### AI Generalist Accelerator Program • Generative AI Business Applications, Outskill (Aug 2026)

Workflow automation (n8n), custom Model Context Protocol (MCP) tooling, agentic architectures, and prompt engineering.

[\[View Certificate ↗\]](#)

### AI Accelerator C13 Hackathon • 2-Day Innovation Challenge, Outskill (Aug 2026)

Rapid prototyping, agent system architecture, and production deployment of generative AI application workflows.

[\[View Award ↗\]](#)



## Education & Academic Background

### Bachelor of Technology (B.Tech) • Computer Science & Engineering

2002 - 2006

Jawaharlal Nehru Technological University (JNTU) • Sri Venkateswara College of Engineering & Technology (SVCET)